

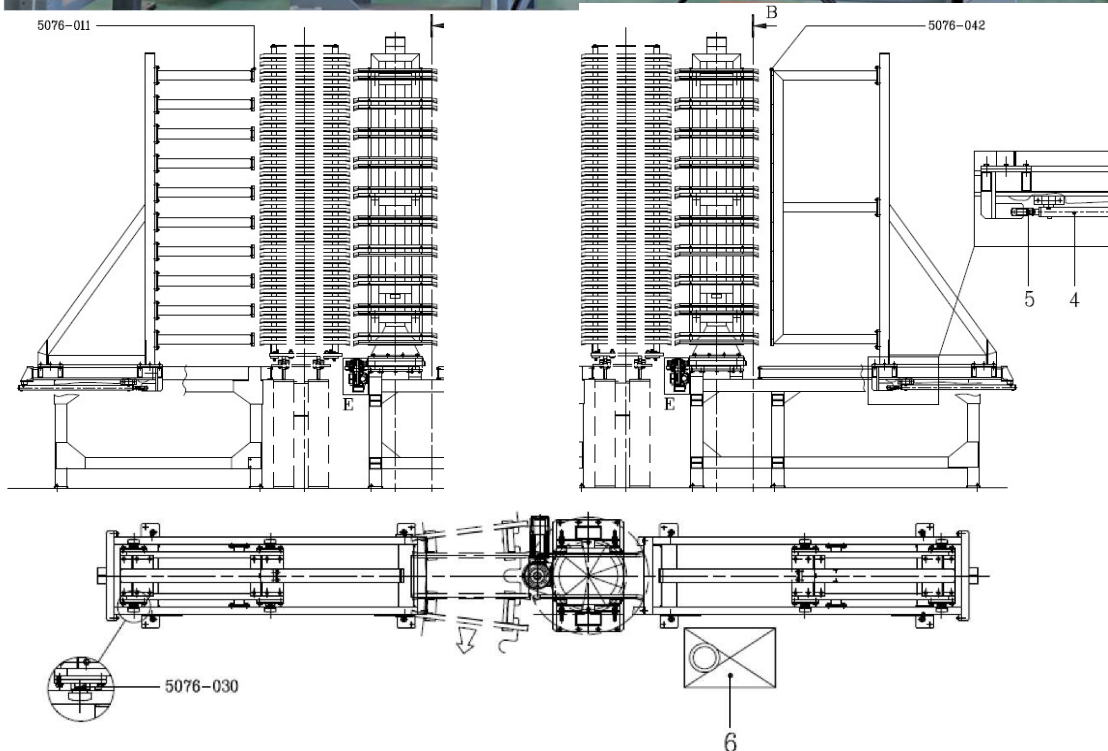
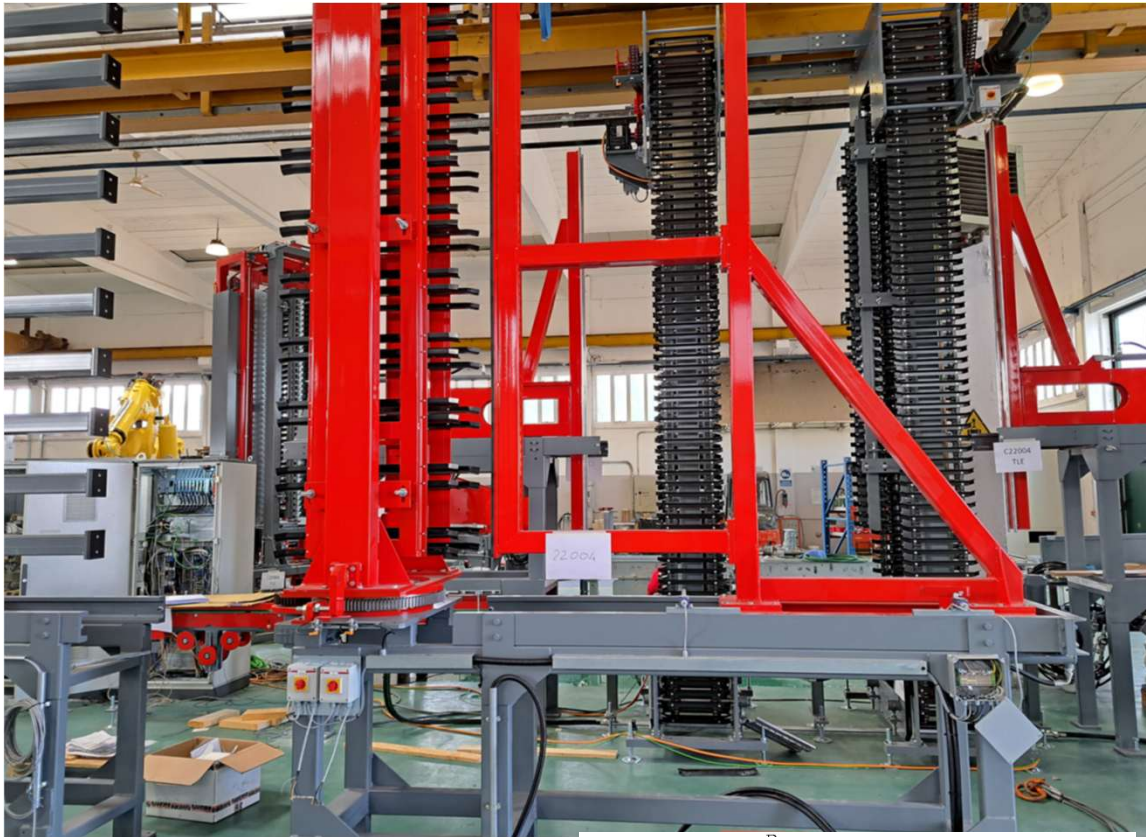
TILES TURNING SYSTEM FOR NOT SELF-STACKING PROFILES-Relationship SOPs

MACHINE TYPE:PK-123 SERIAL NUMBER :M.02238

FP McCANN (Cadeby)

ORDER : C22004

YEAR OF PRODUCTION 2023

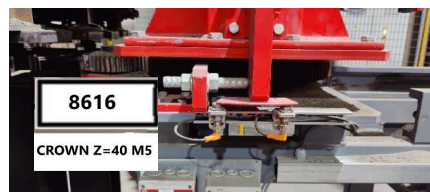
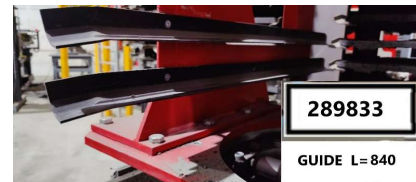
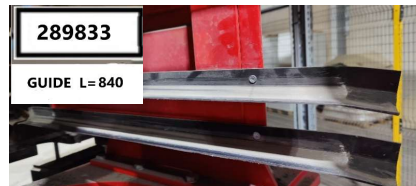


- SOP M.02238.00 Tiles Turning System for not self-stacking profiles-Relationship SOPs
- SOP M.02238.01 180° Rotating Tower.(Tiles Turning System)
- SOP M.02238.02 Rotary Rack internal pusher(Tiles Turning System)
- SOP M.02238.03 Rotary Rack external pusher(Tiles Turning System)
- SOP M.02238.04 Hydraulic Power Unit (Rotary Rack Pushers/Tiles Turning System)

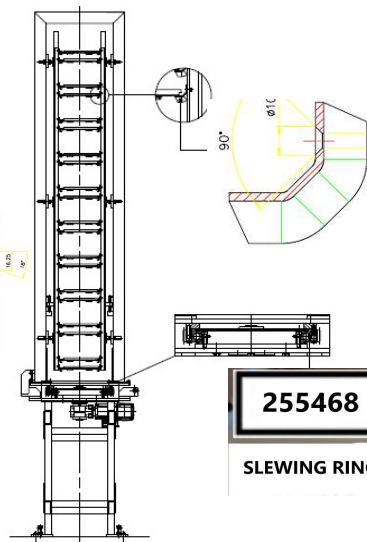
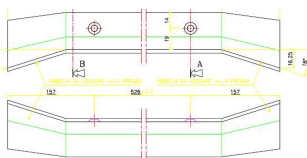
TILES TURNING SYSTEM FOR NOT SELF-STACKING PROFILES

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180° Rotating Tower.(Tiles Turning System)

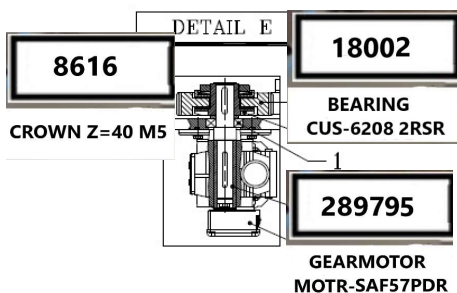


GUIDE L= 840



255468

SLEWING RING



8616

CROWN Z=40 M5

18002

BEARING
CUS-6208 2RSR

289795

GEARMOTOR
MOTR-SAF57PDR

General Description and Function

*This unit is responsible for rotating a set of 20 tiles on a Rotary Rack by 180°. This rotation places them in an opposite and staggered arrangement, allowing two rectangular bundles of 40 tiles to be stacked in a stable and compact manner during the final palletizing stage, optimizing space and load security.

Critical Points and Risk Management

Tower Position Accuracy:

*The exact position at the end of the 180° rotations is critical. Any deviation can cause misplacement of tiles in subsequent guides, leading to jams or damage.

Guide Wear:

*Wear on the guides that direct the tiles during rotation is a primary failure point. Excessive wear can impede smooth and precise rotation, leading to jams or mispositioning.

Drive System Integrity:

*The condition of the crown gear, pinion, and slewing ring is vital for stable, jerk-free rotational movement.

SUPERVISION LEVEL

WEEKLY

8616 Crown Z=40 M5 (1 u)

255468 Slewing Ring (1 u) **MAS INFORMACIÓN.**
Check for dirt and tooth wear. Lubrication.

MONTHLY

252094 Inductive proximity switch (4 units)

289833 Guides L=840 mm (40 units)

18002 Bearing CUS 6208 2RSR (1 u)

289831 Driving Shaft (1)

289795 Gearmotor MOTR-SAF57PDR (1 u)

Verification of wear, alignment, and complete functional state.

Verification of Guides and Positioning:

*Check for wear on the angles of the guides.

*Look for signs of deformation, dents, or excessive abrasion.

*Verify the final necessary rotation position (180°) and calibrate if necessary.

*Check the position and operation of the 4 inductive proximity switches.

*Ensure they are clean, securely mounted, and provide a stable signal for the start/end rotation positions.

Inspection of the Drive System:

Check the gearmotor:

*Verify for abnormal noises, oil leaks, abnormal temperature, and fastener tightness.

*Inspect the condition of the driving shaft and the bearing, looking for radial/axial play or signs of fatigue.

CLEANING

WEEKLY

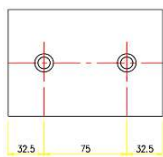
*Support bases, Tower structure, Gearmotor

*Use appropriate tools (brushes, dry cloths) to remove dust, debris, and any accumulated material.

TILES TURNING SYSTEM FOR NOT SELF-STACKING PROFILES

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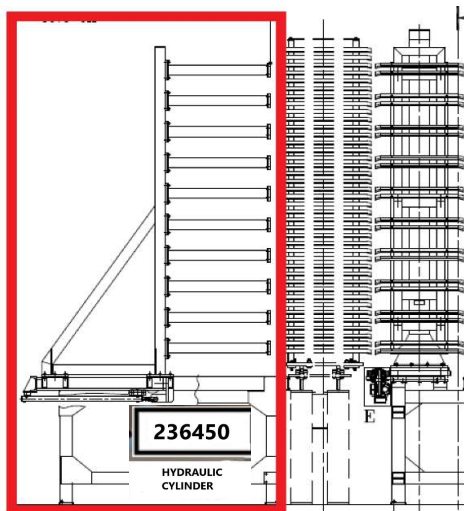
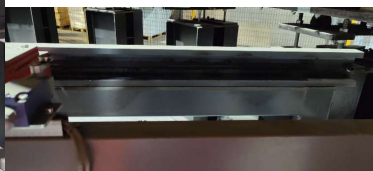
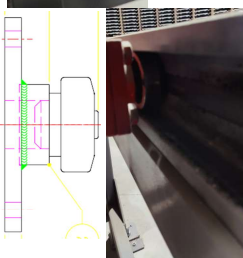
Rotary Rack internal pusher(Tiles Turning System)



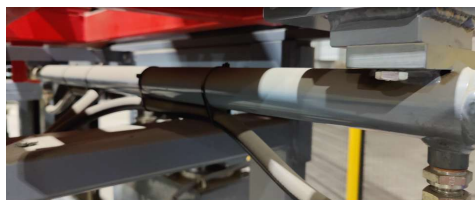
32877

WEAR PLATE

236437

BEARING SUPPORT TROLLEY
(it 277458)

236450

HYDRAULIC
CYLINDER

236450

HYDRAULIC
CYLINDER

General Description and Function

*This system is responsible for shifting the 20 tiles towards the 180° Rotation Tower. It consists of two main parts:

1. A fixed structure equipped with two linear guide
2. A mobile pusher mechanism that travels along these guides, actuated by a single hydraulic

Critical Points and Risk Management

Contact Surface Integrity:

*It is crucial to keep the pusher's contact surfaces clean

Buildup of residue (concrete, dust) can cause:

- *Uneven or misaligned pushing force.
- *Accelerated wear and damage to the tiles.

Travel System Condition:

*The running surfaces and the 4 wheels with bearings on which the pusher structure travels must be free from wear, debris, and damage to ensure smooth and precise movement.

SUPERVISION LEVEL

WEEKLY

32877 Wear Plate Pusher(10 units)

236437 Bearings (it. 27745 (4 units)

Check for wear and alignment.

MONTHLY

236450 Hydraulic Cylinder (1 u)

276168 Fork for pneumatic cylinder (1 u)

18367/77/78 Limit switch (2 units)

WEEKLY INSPECTION & CLEANING

Travel System Inspection

*Check the condition of the 4wheels, bearings, and guides. Look for excessive wear, cracks, or deformation.

Guide & Wheel Cleaning:

*Clean the running surfaces (guides and wheel treads) using a low-pressure air gun (< 3 bar) followed by cloths to remove all dust and abrasive particles.

Wear Plate Pusher Cleaning:

Use steel wire brushes, scrapers, and cloths to remove adhered concrete, paint, resin, or dirt from the wear plate pushers.

*Perform a visual inspection for oil contamination on the floor in the area of the Hydraulic Cylinder , which would indicate a potential leak.

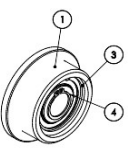
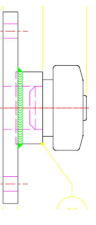
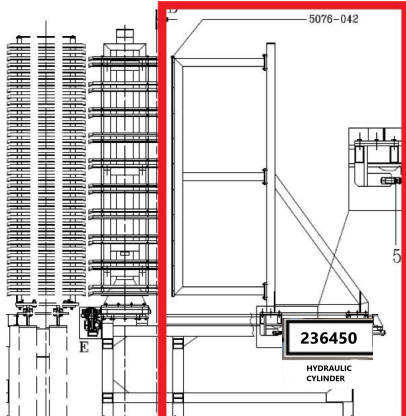
General Components Verification:

*Perform a general check of the condition of all components listed for the week: bearings, fork, and limit switches.

TILES TURNING SYSTEM FOR NOT SELF-STACKING PROFILES

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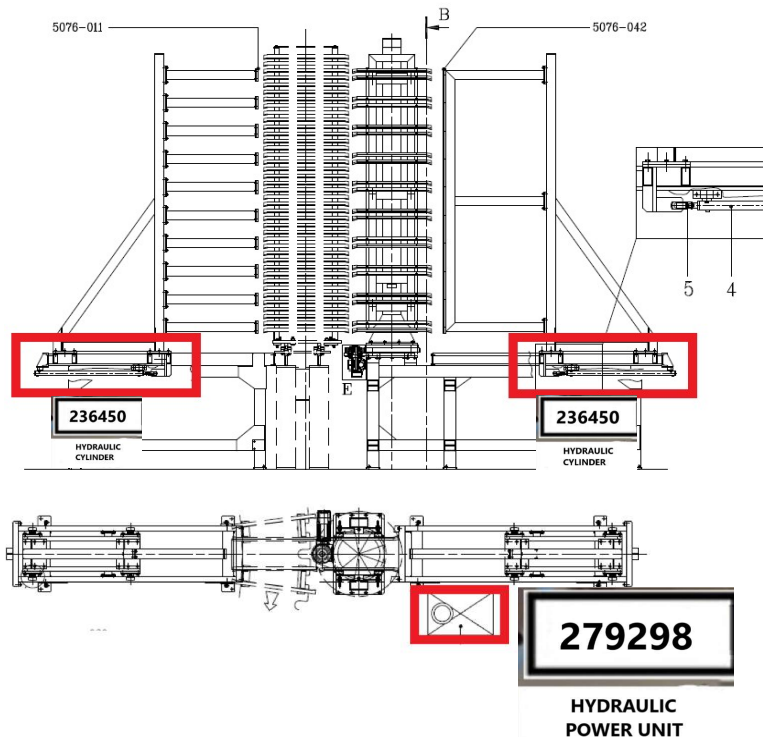
Rotary Rack external pusher(Tiles Turning System)

General Description and Function	
265812 PUSHING PLATE  	
236437 BEARING SUPPORT TROLLEY (it 277458)   	
236450 HYDRAULIC CYLINDER   	
236450 HYDRAULIC CYLINDER  	
SUPERVISION LEVEL	
WEEKLY	
265812 Wear Plate Pusher(1 u) 236437 Bearings (it. 27745 (4 units) Check for wear and alignment.	
MONTHLY	
236450 Hydraulic Cylinder (1 u) 276168 Fork for pneumatic cylinder (1 u) 18367/77/78 Limit switch (2 units)	
WEEKLY INSPECTION & CLEANING	
Travel System Inspection *Check the condition of the 4wheels, bearings, and guides. Look for excessive wear, cracks, or deformation. Guide & Wheel Cleaning: *Clean the running surfaces (guides and wheel treads) using a low-pressure air gun (< 3 bar) followed by cloths to remove all dust and abrasive particles. Wear Plate Pusher Cleaning: Use steel wire brushes, scrapers, and cloths to remove adhered concrete, paint, resin, or dirt from the wear plate pushers. *Perform a visual inspection for oil contamination on the floor in the area of the Hydraulic Cylinder , which would indicate a potential leak. General Components Verification: *Perform a general check of the condition of all components listed for the week: bearings, fork, and limit switches.	

TILES TURNING SYSTEM FOR NOT SELF-STACKING PROFILES

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Hydraulic Power Unit (Rotary Rack Pushers/Tiles Turning System)



General Description and Function

*This system, comprising a hydraulic power unit and 2 hydraulic cylinders, is responsible for pushing the 20 tiles to execute the 180° flip inside the Rotary Rack.

*This step is essential to form the final staggered tile package and achieve a rectilinear, stable arrangement for palletizing.

Critical Points and Risk Management

Leak Control:

*Primary attention must be given to detecting potential oil leaks at the cylinders, connectors, hoses, or solenoid valves of the power unit.

Operational Parameters:

*It is crucial to maintain the specified working pressure and correct oil level in the hydraulic power unit's reservoir.

SHEET PROCESS

Process documentation must include and keep the following reference values updated:

*Working pressure of the hydraulic power unit.

*Oil level to be maintained in the reservoir.

*Advance and retraction times for each pusher, for stop point inside the Rotary Rack -45

SUPERVISION LEVEL

WEEKLY

279298 Hydraulic Power Unit

Solenoid valve (4 units)

General operational check.

Visual Leak Inspection:

* Check the power unit and cylinders for active oil leaks.

Parameter Verification:

*Verify that working pressure and oil level are within specified ranges

MONTHLY

236450 Hydraulic Cylinder (2 units)

276168 Fork for pneumatic cylinder (2 units)

Check fasteners, connectors, and solenoid valves

Mechanical Inspection:

*Verify the tightness and condition of cylinder fasteners and forks.

Hydraulic Inspection:

*Check the integrity of connectors, hydraulic hoses and solenoid valves on the power unit.

Area Inspection:

*Look for oil contamination

CLEANING

WEEKLY

*Surfaces and sides of the hydraulic power unit, contours of the cylinders, and their brackets.

*Use appropriate tools (brushes, dry cloths) to remove dust, debris, and any accumulated material. Prevent dirt from clogging breathers or entering connectors.